

Problem 1. Several stones are placed on an infinite (in both directions) strip of squares. As long as there are at least two stones on a single square, you may pick up two such stones, then move one to the preceding square and one to the following square. Is it possible to return to the starting configuration after a finite sequence of such moves?

Problem 2. Numbers $\frac{49}{1}, \frac{49}{2}, \dots, \frac{49}{97}$ are written on a blackboard. Each time, we can replace two numbers (like a, b) with $2ab - a - b + 1$. After 96 times doing that prenominate action, one number will be left on the board. Find all the possible values for that number.

Problem 3. A circle is divided into six sectors. Then the numbers 1, 0, 1, 0, 0, 0 are written into the sectors. You may increase two neighboring numbers by 1. Is it possible to equalize all numbers by a sequence of such steps?

Problem 4. Given 2004 vertices $a_1, a_2, \dots, a_{2004}$ circling a table in clockwise order. Initially, a_1 is labeled 0 and $a_2, a_3, \dots, a_{2004}$ are labeled 1. In every step, we choose three vertices a_{i-1}, a_i, a_{i+1} with labels a, b, c and change them into $1 - a, 1 - b, 1 - c$ respectively. Is it possible to change all labels into 0?

Problem 5. (USAMO 1994) The sides of a 99-gon are initially colored so that consecutive sides are red, blue, red, blue, $\dots$, red, blue, yellow. We make a sequence of modifications in the coloring, changing the color of one side at a time to one of the three given colors (red, blue, yellow), under the constraint that no two adjacent sides may be the same color. By making a sequence of such modifications, is it possible to arrive at the coloring in which consecutive sides are red, blue, red, blue, red, blue, $\dots$, red, yellow, blue?

Problem 6. (Romanian TST 2002) After elections, every parliament member (PM), has his own absolute rating. When the parliament set up, he enters in a group and gets a relative rating. The relative rating is the ratio of its own absolute rating to the sum of all absolute ratings of the PMs in the group. A PM can move from one group to another only if in his new group his relative rating is greater. In a given day, only one PM can change the group. Show that only a finite number of group moves is possible.

Problem 7. (2022 China TST, Test 1, P3) Let a, b, c, p, q, r be positive integers with $p, q, r \geq 2$. Denote

$$Q = \{(x, y, z) \in \mathbb{Z}^3 : 0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c\}.$$

Initially, some pieces are put on the each point in Q , with a total of M pieces. Then, one can perform the following three types of operations repeatedly: (1) Remove p pieces on (x, y, z) and place a piece on $(x - 1, y, z)$; (2) Remove q pieces on (x, y, z) and place a piece on $(x, y - 1, z)$; (3) Remove r pieces on (x, y, z) and place a piece on $(x, y, z - 1)$.

Find the smallest positive integer M such that one can always perform a sequence of operations, making a piece placed on $(0, 0, 0)$, no matter how the pieces are distributed initially.

Problem 8. (Putnam 2008 A3) Start with a finite sequence $a_1, a_2, \dots, a_n$ of positive integers. If possible, choose two indices $j < k$ such that a_j does not divide a_k and replace a_j and a_k by $\gcd(a_j, a_k)$ and $\text{lcm}(a_j, a_k)$, respectively. Prove that if this process is repeated, it must eventually stop and the final sequence does not depend on the choices made. (Note: $\gcd$ means greatest common divisor and lcm means least common multiple.)

Problem 9. (Korea Final Round 2009) 2008 white stones and 1 black stone are in a row. An 'action' means the following: select one black stone and change the color of neighboring stone(s). Find all possible initial position of the black stone, to make all stones black by finite actions.